

On the Two-Disjoint-Cycle-Cover Pancyclicity of Crossed Cubes*

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Abstract

In this paper, a graph G is two-disjoint-cycle-cover r -pancyclic if for any integer l satisfying $r \leq l \leq |V(G)| - r$, there exist two vertex-disjoint cycles C_1 and C_2 in G such that the lengths of C_1 and C_2 are l and $|V(G)| - l$, respectively, where $|V(G)|$ denotes the total number of vertices in G . Moreover, we define that a graph G is two-disjoint-cycle-cover r -edge-pancyclic if for any two vertex-disjoint edges (u, v) and (x, y) of G , there exist two vertex-disjoint cycles C_1 and C_2 in G such that (i) C_1 contains (u, v) with length l for any integer l satisfying $r \leq l \leq |V(G)| - r$, and (ii) C_2 contains (x, y) with length $|V(G)| - l$. Then, we prove that the n -dimensional crossed cube CQ_n can be two-disjoint-cycle-cover 4-pancyclic and two-disjoint-cycle-cover 2^{n-2} -edge-pancyclic for $n \geq 5$.

1 Introduction

In parallel and distributed computer systems, processors are connected using various types of interconnection networks. In order to guarantee that the system performance can achieve the desired level, various mechanisms of interconnection have been proposed during the last two decades [12, 17, 19]. Typically, the topological structure of an interconnection network is modeled as a graph whose vertices and edges represent processors and communication links, respectively. Among the

many kinds of network topologies, the binary n -cube [18] (for short, hypercube) is one of the most popular networks for parallel and distributed computation. Not only is it ideally suited to both special-purpose and general-purpose tasks, but it can efficiently simulate many other networks [17]. However, the hypercube is bipartite, so it cannot make the best use of network resources. In particular, the hypercube has the largest diameter of the cube family so that it may be disadvantageous to shortest path routing. To compensate for this drawback, many researchers [1, 6, 7, 21] have tried to fashion networks with lower diameters. One such network topology is the crossed cube, which was first proposed by Efe [8]. The crossed cube is derived from the hypercube by changing the connection of some links. Its diameter is about half that of the hypercube [4, 8]. Additionally, the crossed cube has many other attractive properties. For example, it has more cycles than the hypercube [11] and contains binary trees as subgraphs [15]. Moreover, paths of odd and even lengths can be embedded in the crossed cube [5]. The crossed cube has received the attention of many researchers [4, 9, 10, 13, 20] since it was introduced.

Throughout this paper, graphs are finite, simple, and undirected. Some important graph-theory definitions and notations will be introduced first. For those not defined here, we follow the standard terminology given by Bondy and Murty [3]. An *undirected graph* G is a graph with vertex set $V(G)$ and edge set $E(G)$, where $|V(G)| > 0$ and $E(G) \subseteq \{(u, v) | (u, v) \text{ is an unordered pair of elements of } V(G)\}$. Two vertices u and v of G are *adjacent* if $(u, v) \in E(G)$. A graph H is a *subgraph* of G , denoted by $H \subseteq G$, if $V(H) \subseteq V(G)$

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and $E(H) \subseteq E(G)$, and H is a *spanning subgraph* of G (equivalently, H *spans* G) if $V(H) = V(G)$.

A *path* P of length k , $k \geq 1$, from vertex x to vertex y in G is a sequence of distinct vertices $\langle v_1, v_2, \dots, v_{k+1} \rangle$ such that $v_1 = x$, $v_{k+1} = y$, and $(v_i, v_{i+1}) \in E(G)$ for every i , $1 \leq i \leq k$. Moreover, a path of length 0, consisting of a single vertex x , is denoted by $\langle x \rangle$. We can write P as $\langle v_1, v_2, \dots, v_i, Q, v_j, \dots, v_{k+1} \rangle$ for convenience if $Q \subset P$ and $Q = \langle v_i, \dots, v_j \rangle$, where $i \leq j$. The i th vertex of P is denoted by $P(i)$; i.e., $P(i) = v_i$. In particular, let $\text{rev}(P)$ represent the reverse of P ; i.e., $\text{rev}(P) = \langle v_{k+1}, v_k, \dots, v_1 \rangle$. We use $l(P)$ to denote the length of P . The *distance* between two distinct vertices u and v in G , denoted by $d_G(u, v)$, is the length of the shortest path between u and v . A *cycle* is a path with at least three vertices such that the last vertex is adjacent to the first one. For clarity, a cycle of length k , $k \geq 3$, is represented as $\langle v_1, v_2, \dots, v_k, v_1 \rangle$. A path (respectively, cycle) is a *Hamiltonian path* (respectively, *Hamiltonian cycle*) of G if it spans G . We say that G is *Hamiltonian* if it has a Hamiltonian cycle, and G is *Hamiltonian connected* if it contains a Hamiltonian path joining any pair of distinct vertices.

A graph G is called *pancyclic* [2] if it contains a cycle of length l for each integer l from 3 to $|V(G)|$ inclusive. More specifically, a graph G is called *edge-pancyclic* if its any edge lies on a cycle of length l for every l , $3 \leq l \leq |V(G)|$. In this paper, a graph G is *two-disjoint-cycle-cover r -pancyclic* if for any integer l satisfying $r \leq l \leq |V(G)| - r$, there exist two vertex-disjoint cycles C_1 and C_2 in G such that the lengths of C_1 and C_2 are l and $|V(G)| - l$, respectively. Furthermore, we define that a graph G is *two-disjoint-cycle-cover r -edge-pancyclic* if for any two vertex-disjoint edges (u, v) and (x, y) of G , there exist two vertex-disjoint cycles C_1 and C_2 in G such that (i) C_1 contains (u, v) with length l for any integer l satisfying $r \leq l \leq |V(G)| - r$, and (ii) C_2 contains (x, y) with length $|V(G)| - l$. Then, we prove that the n -dimensional crossed cube can be two-disjoint-cycle-cover 4-pancyclic and two-disjoint-cycle-cover 2^{n-2} -edge-pancyclic for $n \geq 5$.

The rest of this paper is organized as follows. In Section 2, we introduce the definition and some topological properties of the crossed cube. The main theorems and their proofs are described in Section 3. Finally, some concluding remarks are given in Section 4.

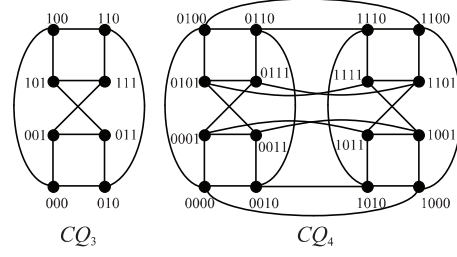


Figure 1: Illustration of CQ_3 and CQ_4 .

2 The crossed cube and its properties

The n -dimensional crossed cube, denoted by CQ_n , has 2^n vertices, each of which corresponds to an n -bit binary string. To define the crossed cube, we need to first introduce an additional concept, “pair related”.

Definition 1. [8] Two 2-bit binary strings $\mathbf{x} = x_2x_1$ and $\mathbf{y} = y_2y_1$ are pair related, denoted by $\mathbf{x} \sim \mathbf{y}$, if and only if $(\mathbf{x}, \mathbf{y}) \in \{(00, 00), (10, 10), (01, 11), (11, 01)\}$.

The formal definition of CQ_n is given below.

Definition 2. [8] The n -dimensional crossed cube CQ_n is recursively constructed as follows:

- (i) CQ_1 is a complete graph with vertex set $\{0, 1\}$.
- (ii) CQ_2 is isomorphic to a cycle of length 4 with vertex set $\{00, 01, 10, 11\}$ and edge set $\{(00, 01), (00, 10), (10, 11), (01, 11)\}$.
- (iii) For $n \geq 3$, let CQ_{n-1}^0 and CQ_{n-1}^1 be two copies of CQ_{n-1} with $V(CQ_{n-1}^0) = \{0u_{n-1}u_{n-2}\dots u_1 \mid u_i = 0 \text{ or } 1 \text{ for } 1 \leq i \leq n-1\}$ and $V(CQ_{n-1}^1) = \{1u_{n-1}u_{n-2}\dots u_1 \mid u_i = 0 \text{ or } 1 \text{ for } 1 \leq i \leq n-1\}$. Then, CQ_n is formed by connecting CQ_{n-1}^0 and CQ_{n-1}^1 with 2^{n-1} edges so that a vertex $\mathbf{u} = 0u_{n-1}u_{n-2}\dots u_1$ of CQ_{n-1}^0 is adjacent to a vertex $\mathbf{v} = 1v_{n-1}v_{n-2}\dots v_1$ of CQ_{n-1}^1 if and only if (1) $u_{n-1} = v_{n-1}$ if n is even, and (2) $u_{2i}u_{2i-1} \sim v_{2i}v_{2i-1}$ for all i , $1 \leq i \leq \lfloor \frac{n-1}{2} \rfloor$.

Figure 1 depicts CQ_3 and CQ_4 . It was proved that CQ_n is n -connected [14] and has diameter $\lceil \frac{n+1}{2} \rceil$ [8].

A vertex $\mathbf{u} = u_nu_{n-1}\dots u_1$ of CQ_n is said to be adjacent to a vertex $\mathbf{v} = v_nv_{n-1}\dots v_1$ along the i th dimension, $1 \leq i \leq n$, if the following four

conditions are all satisfied: (i) $u_i \neq v_i$, (ii) $u_j = v_j$ for all j , $i + 1 \leq j \leq n$, (iii) $u_{2k}u_{2k-1} \sim v_{2k}v_{2k-1}$ for all k , $1 \leq k \leq \lfloor \frac{i-1}{2} \rfloor$, and (iv) $u_{i-1} = v_{i-1}$ if i is even. Then, we say that $\mathbf{u}$ is the i -neighbor of $\mathbf{v}$, denoted by $(\mathbf{v})^i$, and vice versa. The edge $(\mathbf{u}, (\mathbf{u})^i)$ is called an i -dimensional edge. It is easy to see that $\mathbf{v} = (\mathbf{u})^i$ if and only if $\mathbf{u} = (\mathbf{v})^i$.

Chen et al. [5] proposed a method, as the following lemma, for finding a cycle of length 4 with respect to any n -dimensional edge of CQ_n .

Lemma 1. [5] *Let $(\mathbf{u}, \mathbf{v})$ be any n -dimensional edge of CQ_n , $n \geq 3$. For any integer i , $1 \leq i \leq n-1$, vertices $\mathbf{u}$, $\mathbf{v}$, $(\mathbf{u})^i$, and $(\mathbf{v})^i$ induce a cycle of length 4 if i is even or $i = n-1$.*

In [11], Fan et al. described how to locate a cycle of length 5 in CQ_n .

Lemma 2. [11] *Let $(\mathbf{u}, \mathbf{v})$ be any n -dimensional edge of CQ_n , $n \geq 3$. Then, $((\mathbf{u})^1)^n = ((\mathbf{v})^2)^1 = ((\mathbf{v})^1)^2$. Moreover,*

- (i) *vertices $\mathbf{u}$, $\mathbf{v}$, $(\mathbf{u})^1$, $(\mathbf{v})^2$, and $((\mathbf{v})^2)^1$ induce a cycle of length 5, and*
- (ii) *vertices $\mathbf{u}$, $\mathbf{v}$, $(\mathbf{u})^1$, $(\mathbf{v})^1$, and $((\mathbf{v})^1)^2$ induce a cycle of length 5.*

A Hamiltonian graph G is f -fault-tolerant Hamiltonian (respectively, f -fault-tolerant Hamiltonian connected) if $G - F$ remains Hamiltonian (respectively, Hamiltonian connected) for every $F \subseteq V(G) \cup E(G)$ with $|F| \leq f$.

Lemma 3. [13] *For any integer n , $n \geq 3$, CQ_n is $(n-2)$ -fault-tolerant Hamiltonian and $(n-3)$ -fault-tolerant Hamiltonian connected.*

Hereafter, we demonstrate some lemmas used for deriving our main theorems.

Lemma 4. *Let F be a subset of $V(CQ_n)$, where $n \geq 5$ and $0 \leq |F| \leq n$. Besides, let $F_0 = F \cap V(CQ_{n-1}^0)$ and $F_1 = F \cap V(CQ_{n-1}^1)$. If both $CQ_{n-1}^0 - F_0$ and $CQ_{n-1}^1 - F_1$ are Hamiltonian connected, then $CQ_n - F$ is also Hamiltonian connected. Moreover, for any two vertices $\mathbf{u}$ and $\mathbf{v}$ of $CQ_n - F$, there exists a Hamiltonian path P in $CQ_n - F$ joining $\mathbf{u}$ and $\mathbf{v}$ such that P contains a Hamiltonian path of $CQ_{n-1}^0 - F_0$ or a Hamiltonian path of $CQ_{n-1}^1 - F_1$ as its subpath.*

Proof. Without loss of generality, we assume $\mathbf{u} \in V(CQ_{n-1}^0)$. Consider the following two cases.

Case 1. $\mathbf{v} \in V(CQ_{n-1}^0) - F_0$. Let R be a Hamiltonian path of $CQ_{n-1}^0 - F_0$ joining $\mathbf{u}$ and $\mathbf{v}$. For convenience, we write R as $\langle \mathbf{u}, R_1, \mathbf{x}, \mathbf{y}, R_2, \mathbf{v} \rangle$

for some vertices $\mathbf{x}$ and $\mathbf{y}$, where $\{(\mathbf{x})^n, (\mathbf{y})^n\} \cap F_1 = \emptyset$. It is noticed that we can find such $\mathbf{x}$ and $\mathbf{y}$ since $|V(CQ_{n-1}^0) - F_0| \geq 2^{n-1} - n$ and $|F_1| \leq n$. Let S be a Hamiltonian path of $CQ_{n-1}^1 - F_1$ joining $(\mathbf{x})^n$ and $(\mathbf{y})^n$. Then, path $P = \langle \mathbf{u}, R_1, \mathbf{x}, (\mathbf{x})^n, S, (\mathbf{y})^n, \mathbf{y}, R_2, \mathbf{v} \rangle$ is a Hamiltonian path of $CQ_n - F$ joining $\mathbf{u}$ and $\mathbf{v}$ with $S \subset P$.

Case 2. $\mathbf{v} \in V(CQ_{n-1}^1) - F_1$. Let R be a Hamiltonian path of $CQ_{n-1}^0 - F_0$ joining $\mathbf{u}$ and some vertex $\mathbf{z}$ of $CQ_{n-1}^0 - F_0$, where $(\mathbf{z})^n \notin \{\mathbf{v}\} \cup F_1$. Moreover, let S be a Hamiltonian path of $CQ_{n-1}^1 - F_1$ joining $(\mathbf{z})^n$ and $\mathbf{v}$. Then, $P = \langle \mathbf{u}, R, \mathbf{z}, (\mathbf{z})^n, S, \mathbf{v} \rangle$ is a Hamiltonian path of $CQ_n - F$ joining $\mathbf{u}$ and $\mathbf{v}$ with $R \subset P$ and $S \subset P$. $\square$

Lemma 5. [5] *For $n \geq 4$ and $2 \leq i \leq n$, let $(\mathbf{x}, \mathbf{y})$ be any i -dimensional edge of CQ_n . Then, $CQ_n - \{\mathbf{x}, \mathbf{y}\}$ is Hamiltonian connected.*

Lemma 6. *Let $\mathbf{u}$ be any vertex of CQ_n and $(\mathbf{x}, \mathbf{y})$ be any edge of $CQ_n - \{\mathbf{u}\}$ for $n \geq 5$. Then, $CQ_n - \{\mathbf{u}, \mathbf{x}, \mathbf{y}\}$ is Hamiltonian connected.*

Proof. By running a computer program, we can verify that $CQ_5 - \{\mathbf{u}, \mathbf{x}, \mathbf{y}\}$ is Hamiltonian connected. For $n \geq 6$, $CQ_n - \{\mathbf{u}, \mathbf{x}, \mathbf{y}\}$ is Hamiltonian connected by Lemma 3. $\square$

Lemma 7. *Let $(\mathbf{u}, \mathbf{v})$ and $(\mathbf{x}, \mathbf{y})$ be any two vertex-disjoint edges of CQ_n for $n \geq 5$. Then, $CQ_n - \{\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{y}\}$ is Hamiltonian connected.*

Proof. For $n = 5$, we can verify the correctness by brute force with the aid of a computer program. For $n \geq 7$, Lemma 3 states that CQ_n is at least 4-fault-tolerant Hamiltonian connected. Here we need to show the correctness for CQ_6 . Let $F = \{\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{y}\}$. Moreover, let $F_0 = F \cap V(CQ_5^0)$ and $F_1 = F \cap V(CQ_5^1)$. Without loss of generality, assume $|F_0| \geq |F_1|$. Consider the following three cases.

Case 1. $|F_0| = 4$ and $|F_1| = 0$. As mentioned above for $n = 5$, $CQ_5^0 - F_0$ is Hamiltonian connected. Since CQ_5^1 is also Hamiltonian connected, $CQ_6 - F$ is Hamiltonian connected by Lemma 4.

Case 2. $|F_0| = 3$ and $|F_1| = 1$. Obviously, F_0 contains one vertex and two adjacent vertices. By Lemma 6, $CQ_5^0 - F_0$ is Hamiltonian connected. By Lemma 3, $CQ_5^1 - F_1$ is also Hamiltonian connected. Thus, $CQ_6 - F$ is Hamiltonian connected.

Case 3. $|F_0| = 2$ and $|F_1| = 2$. By Lemma 3, both $CQ_5^0 - F_0$ and $CQ_5^1 - F_1$ are Hamiltonian connected. Then, $CQ_6 - F$ is Hamiltonian connected. $\square$

Let $(\mathbf{u}, \mathbf{v})$ be an i -dimensional edge of CQ_n for any i , $1 \leq i \leq n$. Then, $(\mathbf{u}, \mathbf{v})$ is an *even-edge* if i is even.

Lemma 8. *Let F be a subset of $V(CQ_n)$ for $n \geq 5$, where $0 \leq |F| \leq 3$ and F induces a path if $|F| \geq 2$. Then, there exists a Hamiltonian path P in $CQ_n - F$ joining any two vertices and P has at least four vertex-disjoint even-edges.*

Proof. We prove this lemma by induction. With the aid of computer programs, we can verify the validity of the induction base on CQ_5 . The inductive hypothesis is that the statement holds for all CQ_k , $5 \leq k \leq n-1$. We show that the statement also holds for CQ_n . Let $F_0 = F \cap V(CQ_{n-1}^0)$ and $F_1 = F \cap V(CQ_{n-1}^1)$. Without loss of generality, assume $|F_0| \geq |F_1|$. Then, we have the following cases: (1) $|F_0| = 3$ and $|F_1| = 0$, (2) $|F_0| = 2$ and $|F_1| = 1$, (3) $|F_0| = 2$ and $|F_1| = 0$, (4) $|F_0| = 1$ and $|F_1| = 1$, (5) $|F_0| = 1$ and $|F_1| = 0$, and (6) $|F_0| = 0$ and $|F_1| = 0$. In each of the above cases, both $CQ_{n-1}^0 - F_0$ and $CQ_{n-1}^1 - F_1$ are Hamiltonian connected, and $CQ_n - F$ is Hamiltonian connected. By the inductive hypothesis, we have a Hamiltonian path R (respectively, S) joining any two vertices in $CQ_{n-1}^0 - F_0$ (respectively, $CQ_{n-1}^1 - F_1$) that contains at least four vertex-disjoint even-edges. Then, by Lemma 4, there exists a Hamiltonian path P in $CQ_n - F$ joining any two vertices such that P contains R or S as its subpath, and P has at least four vertex-disjoint even-edges. $\square$

The following lemmas discuss some properties about embedding paths of a required length in CQ_n .

Lemma 9. [5] *Let $\mathbf{u}$ and $\mathbf{v}$ be any two vertices of CQ_n for $n \geq 4$. Moreover, let l be any integer with $d_{CQ_n}(\mathbf{u}, \mathbf{v}) \leq l \leq 2^n - 1$ and $l \neq d_{CQ_n}(\mathbf{u}, \mathbf{v}) + 1$. Then, there exists a Hamiltonian path P of CQ_n such that $P(1) = \mathbf{u}$ and $P(l+1) = \mathbf{v}$.*

Lemma 10. [16] *Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{x}$, and $\mathbf{y}$ be any four vertices of CQ_n for $n \geq 5$. Then, there exist two vertex-disjoint paths P_1 and P_2 in CQ_n such that (i) P_1 joins $\mathbf{u}$ and $\mathbf{v}$, (ii) P_2 joins $\mathbf{x}$ and $\mathbf{y}$, and (iii) $l(P_1) = l(P_2) = 2^{n-1} - 1$.*

3 Two-Disjoint-Cycle-Cover Pancyclicity of CQ_n

Theorems 1 and 2 are the main results of this paper.

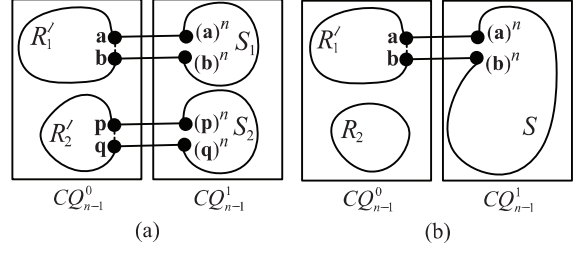


Figure 2: The proof of Theorem 1. (Both dashed and straight lines represent edges.)

Theorem 1. *For $n \geq 5$, CQ_n is two-disjoint-cycle-cover 4-pancyclic.*

Proof. We prove this theorem by induction on n . The validity of the induction base on CQ_5 is verified by brute force with the aid of a computer program. We assume, as the inductive hypothesis, that CQ_{n-1} is two-disjoint-cycle-cover 4-pancyclic. Let l_1 and l_2 be any two integers with $l_1 + l_2 = 2^n$, $l_1 \geq 4$, and $l_2 \geq 4$. Without loss of generality, we assume that $l_1 \geq l_2$; i.e., $l_1 \geq 2^{n-1}$ and $l_2 \leq 2^{n-1}$. Then, we show that there exist two vertex-disjoint cycles C_1 and C_2 of CQ_n such that their lengths are l_1 and l_2 , respectively.

Case 1: $2^{n-2} + 4 \leq l_2 \leq 2^{n-1}$. By the inductive hypothesis, there exist two vertex-disjoint cycles R_1 and R_2 in CQ_{n-1}^0 such that their lengths are $l_2 - 2^{n-2}$ and $2^{n-1} + 2^{n-2} - l_2$, respectively. We can write these two cycles as $R_1 = \langle \mathbf{a}, R_1', \mathbf{b}, \mathbf{a} \rangle$ and $R_2 = \langle \mathbf{p}, R_2', \mathbf{q}, \mathbf{p} \rangle$, respectively, with some vertices $\mathbf{a}$, $\mathbf{b}$, $\mathbf{p}$, and $\mathbf{q}$. By Lemma 10, there exist two vertex-disjoint paths S_1 and S_2 in CQ_{n-1}^1 such that (i) S_1 joins $(\mathbf{b})^n$ and $(\mathbf{a})^n$ with $l(S_1) = 2^{n-2} - 1$, and (ii) S_2 joins $(\mathbf{q})^n$ and $(\mathbf{p})^n$ with $l(S_2) = 2^{n-2} - 1$. Then, $C_1 = \langle \mathbf{a}, R_1', \mathbf{b}, (\mathbf{b})^n, S_1, (\mathbf{a})^n, \mathbf{a} \rangle$ and $C_2 = \langle \mathbf{p}, R_2', \mathbf{q}, (\mathbf{q})^n, S_2, (\mathbf{p})^n, \mathbf{p} \rangle$ are the required cycles. See Figure 2(a) for illustration.

Case 2: $4 \leq l_2 \leq 2^{n-2} + 3$. By the inductive hypothesis, there exist two vertex-disjoint cycles R_1 and R_2 in CQ_{n-1}^0 such that their lengths are $2^{n-1} - l_2$ and l_2 , respectively. We can write R_1 as $\langle \mathbf{a}, R_1', \mathbf{b}, \mathbf{a} \rangle$ with some vertices $\mathbf{a}$ and $\mathbf{b}$. By Lemma 3, there exists a Hamiltonian path S of CQ_{n-1}^1 joining $(\mathbf{b})^n$ and $(\mathbf{a})^n$. Then, $C_1 = \langle \mathbf{a}, R_1', \mathbf{b}, (\mathbf{b})^n, S, (\mathbf{a})^n, \mathbf{a} \rangle$ and $C_2 = R_2$ are the required cycles. See Figure 2(b). $\square$

In our experimental results on CQ_5 , if vertices $\mathbf{u}$, $\mathbf{v}$, $\mathbf{x}$, and $\mathbf{y}$ happen to form a cycle of length 4 with $\mathbf{v} = (\mathbf{u})^2$, $\mathbf{x} = (\mathbf{u})^1$, and $\mathbf{y} = ((\mathbf{u})^1)^2$, then we cannot find any path of length 4 in CQ_5 —

$\{u, v\}$ joining x and y . This implies that CQ_5 is not two-disjoint-cycle-cover 5-edge-pancyclic. Our rudimentary work shows that CQ_n is two-disjoint-cycle-cover 2^{n-2} -edge-pancyclic for $n \geq 5$.

Lemma 11. *Let (u, v) and (x, y) be any two vertex-disjoint edges of CQ_n for $n \geq 6$ such that $\{u, x, y\} \subset V(CQ_{n-1}^0)$. Moreover, let l_1 and l_2 be any two integers with $l_1 + l_2 = 2^n$, $l_1 \geq 2^{n-1}$, and $2^{n-2} \leq l_2 \leq 2^{n-1}$. Suppose that CQ_{n-1} is 2^{n-3} -edge-pancyclic for $n \geq 6$. Then, there exist two vertex-disjoint cycles C_1 and C_2 in CQ_n such that (i) C_1 contains (u, v) with $l(C_1) = l_1$, and (ii) C_2 contains (x, y) with $l(C_2) = l_2$.*

Proof. It suffices to show that there exist two vertex-disjoint paths P_1 and P_2 such that (i) P_1 joins u and v with $l(P_1) = l_1 - 1$, and (ii) P_2 joins x and y with $l(P_2) = l_2 - 1$.

Case 1: $v \in V(CQ_{n-1}^0)$. Two subcases should be considered.

Subcase 1.1: $2^{n-2} + 2^{n-3} \leq l_2 \leq 2^{n-1}$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins u and v with $l(R_1) = 2^{n-1} + 2^{n-2} - l_2 - 1$, and (ii) R_2 joins x and y with $l(R_2) = l_2 - 2^{n-2} - 1$. We can write R_1 and R_2 as $\langle u, R_1^1, a, b, R_1^2, v \rangle$ and $\langle x, R_2^1, p, q, R_2^2, y \rangle$, respectively, for some vertices a, b, p , and q . By Lemma 10, there exist two vertex-disjoint paths S_1 and S_2 in CQ_{n-1}^1 such that (i) S_1 joins $(a)^n$ and $(b)^n$ with $l(S_1) = 2^{n-2} - 1$, and (ii) S_2 joins $(p)^n$ and $(q)^n$ with $l(S_2) = 2^{n-2} - 1$. We set $P_1 = \langle u, R_1^1, a, (a)^n, S_1, (b)^n, b, R_1^2, v \rangle$ and $P_2 = \langle x, R_2^1, p, (p)^n, S_2, (q)^n, q, R_2^2, y \rangle$. Then, $C_1 = \langle u, P_1, v, u \rangle$ and $C_2 = \langle x, P_2, y, x \rangle$ are the required cycles. See Figure 3(a) for illustration.

Subcase 1.2: $2^{n-2} \leq l_2 \leq 2^{n-2} + 2^{n-3} - 1$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins u and v with $l(R_1) = 2^{n-1} - l_2 - 1$, and (ii) R_2 joins x and y with $l(R_2) = l_2 - 1$. We can write R_1 as $\langle u, R_1^1, a, b, R_1^2, v \rangle$ for some vertices a and b . By Lemma 3, there exists a Hamiltonian path S of CQ_{n-1}^1 joining $(a)^n$ and $(b)^n$. We set $P_1 = \langle u, R_1^1, a, (a)^n, S, (b)^n, b, R_1^2, v \rangle$ and $P_2 = R_2$. Then, $C_1 = \langle u, P_1, v, u \rangle$ and $C_2 = \langle x, P_2, y, x \rangle$ are the required cycles. See Figure 3(b) for illustration.

Case 2: $v \in V(CQ_{n-1}^1)$. Let $a \in \{(u)^2, (u)^4, (u)^{n-1}\} - \{x, y\}$.

Subcase 2.1: $2^{n-2} + 2^{n-3} \leq l_2 \leq 2^{n-1}$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, we have two vertex-disjoint paths R_1 and R_2 of CQ_{n-1}^0 such

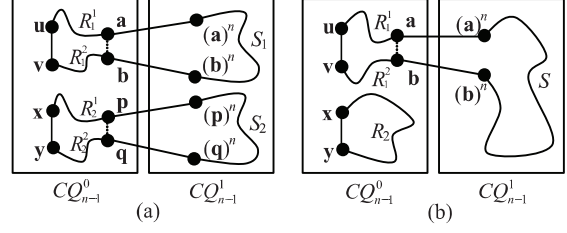


Figure 3: Case 1 in the proof of Lemma 11. (Both dashed and straight lines represent edges.)

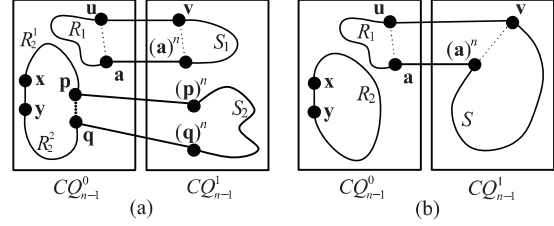


Figure 4: Case 2 in the proof of Lemma 11. (Both dashed and straight lines represent edges.)

that (i) R_1 joins u and a with $l(R_1) = 2^{n-1} + 2^{n-2} - l_2 - 1$, and (ii) R_2 joins x and y with $l(R_2) = l_2 - 2^{n-2} - 1$. Path R_2 can be written as $\langle x, R_2^1, p, q, R_2^2, y \rangle$ for some vertices p and q . By Lemma 10, there exist two vertex-disjoint paths S_1 and S_2 in CQ_{n-1}^1 such that (i) S_1 joins $(a)^n$ and v with $l(S_1) = 2^{n-2} - 1$, and (ii) S_2 joins $(p)^n$ and $(q)^n$ with $l(S_2) = 2^{n-2} - 1$. We set $P_1 = \langle u, R_1, a, (a)^n, S_1, v \rangle$ and $P_2 = \langle x, R_2^1, p, (p)^n, S_2, (q)^n, q, R_2^2, y \rangle$. Then, $C_1 = \langle u, P_1, v, u \rangle$ and $C_2 = \langle x, P_2, y, x \rangle$ are the required cycles. See Figure 4(a).

Subcase 2.2: $2^{n-2} \leq l_2 \leq 2^{n-2} + 2^{n-3} - 1$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, we can find two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins u and a with $l(R_1) = 2^{n-1} - l_2 - 1$, and (ii) R_2 joins x and y with $l(R_2) = l_2 - 1$. Since there exists a Hamiltonian path S in CQ_{n-1}^1 joining $(a)^n$ and v , $C_1 = \langle u, R_1, a, (a)^n, S, v, u \rangle$ and $C_2 = \langle x, R_2, y, x \rangle$ are the required cycles. See Figure 4(b). $\square$

Lemma 12. *Let (u, v) and (x, y) be any two vertex-disjoint edges of CQ_n for $n \geq 6$ such that $u \in V(CQ_{n-1}^0)$, $y \in V(CQ_{n-1}^1)$, and $|\{v, x\} \cap V(CQ_{n-1}^0)| \geq 1$. Moreover, let l_1 and l_2 be any two integers with $l_1 + l_2 = 2^n$, $l_1 \geq 2^{n-1}$, and $2^{n-2} \leq l_2 \leq 2^{n-1}$. Suppose that CQ_{n-1} is 2^{n-3} -*

edge-pancyclic for $n \geq 6$. Then, there exist two vertex-disjoint cycles C_1 and C_2 in CQ_n such that (i) C_1 contains $(\mathbf{u}, \mathbf{v})$ with $l(C_1) = l_1$, and (ii) C_2 contains $(\mathbf{x}, \mathbf{y})$ with $l(C_2) = l_2$.

Proof. It suffices to show that there exist two vertex-disjoint paths P_1 and P_2 such that (i) P_1 joins $\mathbf{u}$ and $\mathbf{v}$ with $l(P_1) = l_1 - 1$, and (ii) P_2 joins $\mathbf{x}$ and $\mathbf{y}$ with $l(P_2) = l_2 - 1$.

Case 1: $\mathbf{v} \in V(CQ_{n-1}^0)$ and $\mathbf{x} \in V(CQ_{n-1}^1)$. The following subcases are distinguished.

Subcase 1.1: $l_2 = 2^{n-1}$. Obviously, we have a Hamiltonian path P_1 of CQ_{n-1}^0 joining $\mathbf{u}$ and $\mathbf{v}$ and a Hamiltonian path P_2 of CQ_{n-1}^1 joining $\mathbf{x}$ and $\mathbf{y}$. Then, $C_1 = \langle \mathbf{u}, P_1, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, P_2, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles.

Subcase 1.2: $l_2 = 2^{n-1} - 1$. Since $n \geq 6$, by Lemma 2, we can find a neighbor $\mathbf{a}$ of $\mathbf{u}$ in CQ_{n-1}^0 such that $\{\mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1\}$ induces a cycle of length 5 and $\{\mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1\} \cap \{\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{y}\} = \emptyset$, where $\mathbf{b} = (\mathbf{a})^n$. By Lemma 8, there exists a Hamiltonian path S in $CQ_{n-1}^1 - \{\mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2\}$ joining $\mathbf{x}$ and $\mathbf{y}$ such that S has at least four vertex-disjoint even-edges. Among these even-edges, we can find one even-edge $(\mathbf{p}, \mathbf{q})$ such that $\{(\mathbf{p})^n, (\mathbf{q})^n\} \cap \{\mathbf{u}, \mathbf{v}\} = \emptyset$. It is noticed that $((\mathbf{p})^n, (\mathbf{q})^n) \in E(CQ_n)$. Path S can be written as $\langle \mathbf{x}, S_1, \mathbf{p}, \mathbf{q}, S_2, \mathbf{y} \rangle$. By Lemma 7, there exists a Hamiltonian path R in $CQ_{n-1}^0 - \{\mathbf{u}, \mathbf{a}, (\mathbf{p})^n, (\mathbf{q})^n\}$ joining $(\mathbf{a})^1$ and $\mathbf{v}$. Then, $C_1 = \langle \mathbf{u}, \mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1, R, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S_1, \mathbf{p}, (\mathbf{p})^n, (\mathbf{q})^n, \mathbf{q}, S_2, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 5(a).

Subcase 1.3: $l_2 = 2^{n-1} - 2$. Let R be a Hamiltonian path of CQ_{n-1}^0 joining $\mathbf{u}$ and $\mathbf{v}$. By Lemma 8, there exists an even-edge $(\mathbf{a}, \mathbf{b})$ of R such that $\{(\mathbf{a})^n, (\mathbf{b})^n\} \cap \{\mathbf{x}, \mathbf{y}\} = \emptyset$. We can write R as $\langle \mathbf{u}, R_1, \mathbf{a}, \mathbf{b}, R_2, \mathbf{v} \rangle$. By Lemma 3, there exists a Hamiltonian path S in $CQ_{n-1}^1 - \{(\mathbf{a})^n, (\mathbf{b})^n\}$ joining $\mathbf{x}$ and $\mathbf{y}$. Then, $C_1 = \langle \mathbf{u}, R_1, \mathbf{a}, (\mathbf{a})^n, (\mathbf{b})^n, \mathbf{b}, R_2, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 5(b).

Subcase 1.4: $l_2 = 2^{n-1} - 3$. As described in Subcase 1.2, we can find a neighbor $\mathbf{a}$ of $\mathbf{u}$ in CQ_{n-1}^0 such that $\{\mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1\}$ induces a cycle of length 5 and $\{\mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1\} \cap \{\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{y}\} = \emptyset$, where $\mathbf{b} = (\mathbf{a})^n$. By Lemma 3, there exists a Hamiltonian path R in $CQ_{n-1}^0 - \{\mathbf{u}, \mathbf{a}\}$ joining $(\mathbf{a})^1$ and $\mathbf{v}$. By Lemma 6, there exists a Hamiltonian path S in $CQ_{n-1}^1 - \{\mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2\}$ joining $\mathbf{x}$ and $\mathbf{y}$. Then, $C_1 = \langle \mathbf{u}, \mathbf{a}, \mathbf{b}, (\mathbf{b})^1, ((\mathbf{b})^1)^2, (\mathbf{a})^1, R, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S, \mathbf{y}, \mathbf{x} \rangle$ form the required cycles. See Figure 5(c).

Subcase 1.5: $2^{n-2} \leq l_2 \leq 2^{n-1} - 4$. By Lemma 9, we can find a Hamiltonian path S of CQ_{n-1}^1 such that $S(1) = \mathbf{x}$ and $S(l_2) = \mathbf{y}$. Path S can be written as $\langle \mathbf{x}, S_1, \mathbf{y}, \mathbf{a}, S_2, \mathbf{b} \rangle$ for some vertices $\mathbf{a}$ and $\mathbf{b}$, where $\mathbf{a}$ is a neighbor of $\mathbf{y}$ on S . The following conditions are distinguished.

Condition 1.5.1: $\{\mathbf{u}, \mathbf{v}\} \cap \{(\mathbf{a})^n, (\mathbf{b})^n\} = \emptyset$. By Lemma 10, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins $\mathbf{u}$ and $(\mathbf{a})^n$ with $l(R_1) = 2^{n-2} - 1$, and (ii) R_2 joins $\mathbf{v}$ and $(\mathbf{b})^n$ with $l(R_2) = 2^{n-2} - 1$. Then, $C_1 = \langle \mathbf{u}, R_1, (\mathbf{a})^n, \mathbf{a}, S_2, \mathbf{b}, (\mathbf{b})^n, R_2, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S_1, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 5(d).

Condition 1.5.2: $\mathbf{u} = (\mathbf{a})^n$ and $\mathbf{v} \neq (\mathbf{b})^n$. By Lemma 3, we have a Hamiltonian path R of $CQ_{n-1}^0 - \{\mathbf{u}\}$ joining $\mathbf{v}$ and $(\mathbf{b})^n$. Then, $C_1 = \langle \mathbf{u}, \mathbf{a}, S_2, \mathbf{b}, (\mathbf{b})^n, R, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S_1, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 5(e).

Condition 1.5.3: $\mathbf{u} \neq (\mathbf{a})^n$ and $\mathbf{v} = (\mathbf{b})^n$. Similarly, since there exists a Hamiltonian path R in $CQ_{n-1}^0 - \{\mathbf{v}\}$ joining $\mathbf{u}$ and $(\mathbf{a})^n$, we can set $C_1 = \langle \mathbf{u}, R, (\mathbf{a})^n, \mathbf{a}, S_2, \mathbf{b}, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S_1, \mathbf{y}, \mathbf{x} \rangle$ as the required cycles.

Condition 1.5.4: $\{\mathbf{u}, \mathbf{v}\} = \{(\mathbf{a})^n, (\mathbf{b})^n\}$. For the sake of simplicity, we write S_2 as $\langle \mathbf{a}, S_2^1, \mathbf{p}, \mathbf{q}, S_2^2, \mathbf{b} \rangle$ with some vertices $\mathbf{p}$ and $\mathbf{q}$. Obviously, there exists a Hamiltonian path R in $CQ_{n-1}^0 - \{\mathbf{u}, \mathbf{v}\}$ joining $(\mathbf{p})^n$ and $(\mathbf{q})^n$. Then, $C_1 = \langle \mathbf{u}, \mathbf{a}, S_2^1, \mathbf{p}, (\mathbf{p})^n, R, (\mathbf{q})^n, \mathbf{q}, S_2^2, \mathbf{b}, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, S_1, \mathbf{y}, \mathbf{x} \rangle$ form the required cycles. Figure 5(f) illustrates this condition.

Case 2: $\{\mathbf{v}, \mathbf{x}\} \subset V(CQ_{n-1}^0)$. Let $\mathbf{p} \in \{(\mathbf{x})^2, (\mathbf{x})^4, (\mathbf{x})^{n-1}\} - \{\mathbf{u}, \mathbf{v}\}$. Consider the following subcases.

Subcase 2.1: $2^{n-2} + 2^{n-3} + 3 \leq l_2 \leq 2^{n-1}$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins $\mathbf{u}$ and $\mathbf{v}$ with $l(R_1) = 2^{n-1} + 2^{n-2} - l_2 - 1$, and (ii) R_2 joins $\mathbf{x}$ and $\mathbf{p}$ with $l(R_2) = l_2 - 2^{n-2} - 1$. We can write R_1 as $\langle \mathbf{u}, R_1^1, \mathbf{a}, \mathbf{b}, R_1^2, \mathbf{v} \rangle$ for some vertices $\mathbf{a}$ and $\mathbf{b}$. By Lemma 10, there exist two vertex-disjoint paths S_1 and S_2 in CQ_{n-1}^1 such that (i) S_1 joins $(\mathbf{a})^n$ and $(\mathbf{b})^n$ with $l(S_1) = 2^{n-2} - 1$, and (ii) S_2 joins $(\mathbf{p})^n$ and $\mathbf{y}$ with $l(S_2) = 2^{n-2} - 1$. Then, $C_1 = \langle \mathbf{u}, R_1^1, \mathbf{a}, (\mathbf{a})^n, S_1, (\mathbf{b})^n, \mathbf{b}, R_1^2, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, R_2, \mathbf{p}, (\mathbf{p})^n, S_2, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 6(a).

Subcase 2.2: $2^{n-2} \leq l_2 \leq 2^{n-2} + 2^{n-3} + 2$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins $\mathbf{u}$ and $\mathbf{v}$ with $l(R_1) = 2^{n-1} - l_2 + 1$, and (ii) R_2 joins $\mathbf{x}$ and

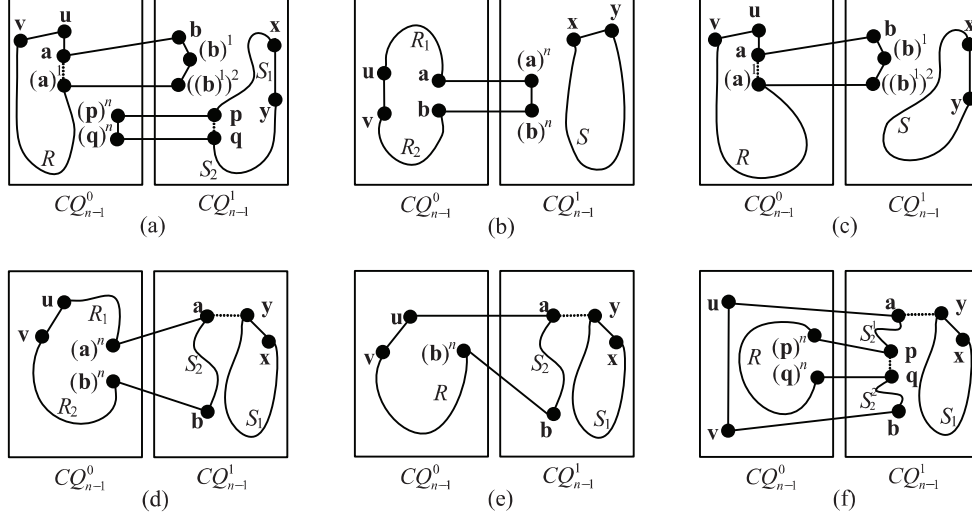


Figure 5: Case 1 in the proof of Lemma 12. (Both dashed and straight lines represent edges.)

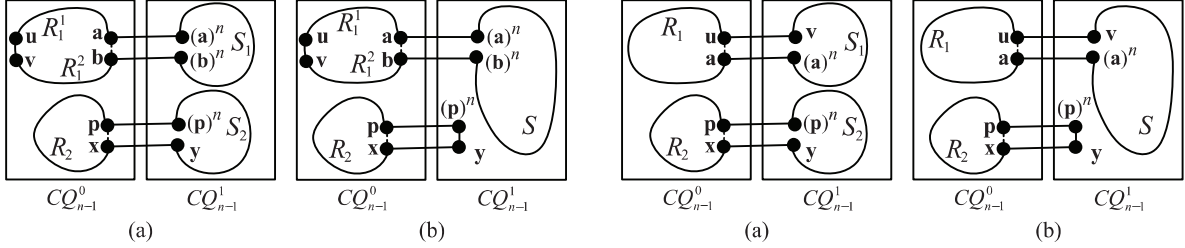


Figure 6: Case 2 in the proof of Lemma 12. (Both dashed and straight lines represent edges.)

$\mathbf{p}$ with $l(R_2) = l_2 - 3$. We can write R_1 as $\langle \mathbf{u}, R_1^1, \mathbf{a}, \mathbf{b}, R_1^2, \mathbf{v} \rangle$ for some vertices $\mathbf{a}$ and $\mathbf{b}$. By Lemma 3, there exists a Hamiltonian path S of $CQ_{n-1}^1 - \{(\mathbf{p})^n, \mathbf{y}\}$ joining $(\mathbf{a})^n$ and $(\mathbf{b})^n$. Then, $C_1 = \langle \mathbf{u}, R_1^1, \mathbf{a}, (\mathbf{a})^n, S, (\mathbf{b})^n, \mathbf{b}, R_1^2, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, R_2, \mathbf{p}, (\mathbf{p})^n, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 6(b).

Case 3: $\mathbf{x} \in V(CQ_{n-1}^0)$ and $\mathbf{v} \in V(CQ_{n-1}^1)$. Let $\mathbf{a} \in \{(\mathbf{u})^2, (\mathbf{u})^4, (\mathbf{u})^{n-1}\} - \{\mathbf{x}\}$ and $\mathbf{p} \in \{(\mathbf{x})^2, (\mathbf{x})^4, (\mathbf{x})^{n-1}\} - \{\mathbf{u}\}$ such that $\mathbf{a} \neq \mathbf{p}$.

Subcase 3.1: $2^{n-2} + 2^{n-3} + 3 \leq l_2 \leq 2^{n-1}$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins $\mathbf{u}$ and $\mathbf{a}$ with $l(R_1) = 2^{n-1} + 2^{n-2} - l_2 - 1$, and (ii) R_2 joins $\mathbf{x}$ and $\mathbf{p}$ with $l(R_2) = l_2 - 2^{n-2} - 1$. By Lemma 10, there exist two vertex-disjoint paths S_1 and S_2 in CQ_{n-1}^1 such that (i) S_1 joins $(\mathbf{a})^n$ and $\mathbf{v}$ with $l(S_1) = 2^{n-2} - 1$, and (ii) S_2 joins $(\mathbf{p})^n$ and $\mathbf{y}$ with $l(S_2) = 2^{n-2} - 1$.

Figure 7: Case 3 in the proof of Lemma 12. (Both dashed and straight lines represent edges.)

Then, $C_1 = \langle \mathbf{u}, R_1, \mathbf{a}, (\mathbf{a})^n, S_1, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, R_2, \mathbf{p}, (\mathbf{p})^n, S_2, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 7(a).

Subcase 3.2: $2^{n-2} \leq l_2 \leq 2^{n-2} + 2^{n-3} + 2$. Since CQ_{n-1} is 2^{n-3} -edge-pancyclic, there exist two vertex-disjoint paths R_1 and R_2 in CQ_{n-1}^0 such that (i) R_1 joins $\mathbf{u}$ and $\mathbf{a}$ with $l(R_1) = 2^{n-1} - l_2 + 1$, and (ii) R_2 joins $\mathbf{x}$ and $\mathbf{p}$ with $l(R_2) = l_2 - 3$. By Lemma 3, there exists a Hamiltonian path S of $CQ_{n-1}^1 - \{(\mathbf{p})^n, \mathbf{y}\}$ joining $(\mathbf{a})^n$ and $\mathbf{v}$. Then, $C_1 = \langle \mathbf{u}, R_1, \mathbf{a}, (\mathbf{a})^n, S, \mathbf{v}, \mathbf{u} \rangle$ and $C_2 = \langle \mathbf{x}, R_2, \mathbf{p}, (\mathbf{p})^n, \mathbf{y}, \mathbf{x} \rangle$ are the required cycles. See Figure 7(b). $\square$

With the aid of a computer program, we verified by brute force CQ_5 is two-disjoint-cycle-cover 8-edge-pancyclic. Using both Lemmas 11 and 12 we further derive the following theorem.

Theorem 2. For $n \geq 5$, CQ_n is two-disjoint-cycle-cover 2^{n-2} -edge-pancyclic.

4 Concluding remarks

In this paper, we first prove that CQ_n is two-disjoint-cycle-cover 4-pancyclic. However, CQ_n is not two-disjoint-cycle-cover 5-edge-pancyclic. The search result indicates that there exists no path of length 4 in $CQ_5 - \{\mathbf{u}, \mathbf{v}\}$ joining $\mathbf{x}$ and $\mathbf{y}$ if $\mathbf{u}, \mathbf{v}, \mathbf{x}$, and $\mathbf{y}$ happen to form a cycle of length 4 with some conditions such as (i) $\mathbf{v} = (\mathbf{u})^2$, $\mathbf{x} = (\mathbf{u})^1$, and $\mathbf{y} = ((\mathbf{u})^1)^2$, and (ii) $\mathbf{v} = ((\mathbf{u})^1)^2$, $\mathbf{x} = (\mathbf{u})^1$, and $\mathbf{y} = (\mathbf{u})^2$. Therefore, our rudimentary work presents that the crossed cube CQ_n can be two-disjoint-cycle-cover 2^{n-2} -edge-pancyclic for $n \geq 5$.

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